

point. The three forces should form a closed triangle.

6

C

$$U_{\text{balloon}} + U_{\text{block}} = kx + W$$

$$(0.015)(1000)g + (0.005)(1000)g = 5000x + 60$$

$$x = 0.0272 \text{ m}$$

7

D

$$\sqrt{m_1 m_2}$$